

Tao2.3

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1 Uniform Convergence

Let X, Y be metric spaces. Suppose (f_n) converges uniformly to f . Then (f_n) converges point-wise to f .

Proof: Let $x_0 \in X$. Let $\epsilon > 0$. Since (f_n) converges uniformly to f , there exists $N \in \mathbb{N}$ st $n \geq N \rightarrow d_Y(f_n(x), f(x)) < \epsilon$ for all $n \geq N$. In particular, $d_Y(f_n(x_0), f(x_0)) < \epsilon$ for all $n \geq N$, so $\lim_{n \rightarrow \infty} f_n(x_0) = f(x_0)$.

Let X, Y be metric spaces, $E \subset X$ and $x_0 \in \overline{E}$. Suppose $f_n : E \rightarrow Y$ such that $\lim_{x \rightarrow x_0; x \in E} f_n(x)$ exists and that the functions f_n converge uniformly to f . Then $\lim_{n \rightarrow \infty} \lim_{x \rightarrow x_0; x \in E} f_n(x)$ and $\lim_{x \rightarrow x_0; x \in E} f(x)$ both exist and equal each other.

Proof: Let $L_n = \lim_{x \rightarrow x_0; x \in E} f_n(x)$. Since (f_n) is convergent, (f_n) is Cauchy. Let $\epsilon > 0$ and pick $N \in \mathbb{N}$ such that $n, m \geq N \rightarrow \rho(f_n, f_m) < \epsilon$. Fix n, m . There exists $\delta > 0$ st $x \in V_\delta(x_0) \cap E$ implies $d_Y(f_n(x), L_n) < \epsilon$ and $d_Y(f_m(x), L_m) < \epsilon$. Pick any $x_1 \in V_\delta(x_0) \cap E$ (nonempty since $x_0 \in \overline{E}$), so

$$\begin{aligned} d_Y(L_n, L_m) &\leq d_Y(L_n, f_n(x_1)) + d_Y(f_n(x_1), f_m(x_1)) + d_Y(f_m(x_1), L_m) \\ &< \epsilon + \epsilon + \epsilon = 3\epsilon. \end{aligned}$$

It follows that the sequence (L_n) is Cauchy in Y . Since Y is complete, let L be the limit.

Let $\epsilon > 0$. There exists $n \in \mathbb{N}$ such that $\rho(f_n, f) < \epsilon$ and $d_Y(L_n, L) < \epsilon$. Let $\delta > 0$ such that $x \in V_\delta(x_0) \cap E \rightarrow d_Y(f_n(x), L_n) < \epsilon$. Then for $x \in V_\delta(x_0) \cap E$

$$\begin{aligned} d_Y(f(x), L) &\leq d_Y(f(x), f_n(x)) + d_Y(f_n(x), L_n) + d_Y(L_n, L) \\ &< \epsilon + \epsilon + \epsilon = 3\epsilon, \end{aligned}$$

so $\lim_{x \rightarrow x_0; x \in E} f(x) = L$.

Suppose we have a sequence of continuous functions $f_n : X \rightarrow Y$ converging

uniformly to $f : X \rightarrow Y$. Let $x_0 \in X$ and (x_n) converge to x_0 . Then

$$\lim_{n \rightarrow \infty} f_n(x_n) = f(x).$$

Proof: Let $\epsilon > 0$. Since f is continuous, pick $\delta > 0$ such that $d_X(x, x_0) < \delta \rightarrow d_Y(f(x), f(x_0)) < \epsilon$. Since (f_n) converges uniformly to f and (x_n) converges to x_0 , pick $N \in \mathbb{N}$ such that $n \geq N$ implies $d_X(x_n, x_0) < \delta$ and $\rho(f, f_n) < \epsilon$. For $n \geq N$,

$$\begin{aligned} d_Y(f_n(x_n), f(x)) &\leq d_Y(f_n(x_n), f(x_n)) + d_Y(f(x_n), f(x)) \\ &< \epsilon + \epsilon = 2\epsilon. \end{aligned}$$

Suppose (f_n) converges uniformly to f . If each f_n is bounded, so is f .

Proof: Since (f_n) is convergent, (f_n) is Cauchy. Pick N st $n, m \geq N$ implies $\rho(f_n, f_m) < 1$. Let $f_N(X) \subset B(y_N, R_N)$. Then for all $x \in X$ and $k \geq N$,

$$\begin{aligned} d_Y(f_k(x), y_N) &\leq d_Y(f_k(x), f_N(x)) + d_Y(f_N(x), y_N) \\ &< 1 + R_N. \end{aligned}$$

Suppose for $k \in [N-1]$ that $f_k(X) \subset B(y_k, R_k)$.

Let X be a metric space. Then $B(x_1, R_1) \cup B(x_2, R_2)$ is bounded.

Proof: For $x \in B(x_2, R_2)$,

$$d_X(x, x_1) \leq d_X(x, x_2) + d_X(x_2, x_1) < R_2 + d_X(x_2, x_1).$$

Then $B(x_1, R_1) \cup B(x_2, R_2) \subset B(x_1, \max(R_1, R_2 + d_X(x_1, x_2)))$.

By induction, $\bigcup_{k \in [N-1]} B(y_k, R_k)$ is bounded and contains the images of $f_1, f_2, \dots, f_{N-1}$. The images of $f_N, f_{N+1}, \dots$ are also contained in a bounded set, so the images of $f_1, f_2, \dots$ are contained in a bounded set. f is also the point-wise limit of the f_n , so f is contained in a bounded set (the limit of a sequence contained in a bounded set is contained in the closure of that bounded set, which is bounded).

Let X be a metric space, $f_n, g_n : X \rightarrow \mathbb{R}$ converge uniformly to f, g , resp and both be uniformly bdd by M . Then $f_n g_n$ converges uniformly to fg .

Let $\epsilon > 0$. Pick $N \in \mathbb{N}$ such that $\rho(f_n, f) < \epsilon$ and $\rho(g_n, g) < \epsilon$ for $n \geq N$. For

$n \geq N$,

$$\begin{aligned}
\rho(f_n g_n, f g) &= \sup_{x \in X} |f_n(x) g_n(x) - f(x) g(x)| \\
&\leq \sup_{x \in X} |f_n(x) g_n(x) - f_n(x) g(x)| + |f_n(x) g(x) - f(x) g(x)| \\
&\leq \sup_{x \in X} |f_n(x)| \cdot |g_n(x) - g(x)| + \sup_{x \in X} |g(x)| \cdot |f_n(x) - f(x)| \\
&\leq M\epsilon + M\epsilon = 2M\epsilon.
\end{aligned}$$

Since $2M$ is fixed, $f_n g_n \rightarrow f g$ uniformly, as desired.

2 The Metric of Uniform Convergence

Let X, Y be metric spaces with $X \neq \emptyset$. Let $B(X \rightarrow Y) = \{f \in Y^X : f \text{ is bounded}\}$. Define $d_\infty : B(X \rightarrow Y) \times B(X \rightarrow Y) \rightarrow \mathbb{R}$ by $d_\infty(f, g) = \sup_{x \in X} d_Y(f(x), g(x))$. Let $f(X) \subset B(y_0, R_0)$ and $g(X) \subset B(y_1, R_1)$. Then for all $x \in X$,

$$\begin{aligned}
d_Y(f(x), g(x)) &\leq d_Y(f(x), y_0) + d_Y(y_0, y_1) + d_Y(y_1, g(x)) \\
&< R_0 + d_Y(y_0, y_1) + R_1.
\end{aligned}$$

Since the RHS is constant and $\{d_Y(f(x), g(x)) : x \in X\} \neq \emptyset$, d_∞ is well-defined.

$$\begin{aligned}
d_\infty(f, g) + d_\infty(g, h) &= \sup_{x \in X} d_Y(f(x), g(x)) + \sup_{x \in X} d_Y(g(x), h(x)) \\
&\geq d_Y(f(x), g(x)) + d_Y(g(x), h(x)) \\
&\geq d_Y(f(x), h(x)).
\end{aligned}$$

Since this holds for all $x \in X$, $d_\infty(f, h) \leq d_\infty(f, g) + d_\infty(g, h)$. d_∞ is symmetric.

Suppose $d_\infty(f, g) = 0$. Then $d_Y(f(x), g(x)) = 0$ for all $x \in X$, so $f(x) = g(x)$ for all $x \in X$, ie $f = g$.

If Y is complete, show that $C(X \rightarrow Y)$ is complete.

Suppose (f_n) is Cauchy. Fix $x_0 \in X$, so $(f_n(x_0))$ is Cauchy. Since Y is complete, $(f_n(x_0))$ is convergent, so let $f_n(x_0) \rightarrow f(x_0)$.

Let $\epsilon > 0$. Pick $N \in \mathbb{N}$ st $n, m \geq N \rightarrow d_\infty(f_n, f_m) < \epsilon$. Fix $x_0 \in X$, so $d_Y(f_n(x_0), f_m(x_0)) < \epsilon$, ie $f_m(x_0) \in B(f_n(x_0), \epsilon)$. Let $m \rightarrow \infty$, so $f(x_0) \in \overline{B(f_n(x_0), \epsilon)} \subset D(f_n(x_0), \epsilon)$, ie $d_Y(f(x_0), f_n(x_0)) \leq \epsilon$, so $d_\infty(f, f_n) \leq \epsilon$ for all $n \geq N$, ie (f_n) converges uniformly to f .

Since f_n is bdd, so is f . Since f_n is continuous, so is f , so $f \in C(X \rightarrow Y)$

and $C(X \rightarrow Y)$ is complete.

3 Weierstrass M-Test

Let $f_n \in C(X \rightarrow \mathbb{R})$ st $\sum_{n=1}^{\infty} \|f_n\|_{\infty}$ converges. Then $\sum_{n=1}^{\infty} f_n$ converges uniformly to some $f \in C(X \rightarrow \mathbb{R})$.

Proof: For any $n \in N$, $\sum_{k=1}^n f_k \in C(X \rightarrow \mathbb{R})$. Let $\epsilon > 0$. By tail test, pick $N \in N$ st $n, m \geq N \rightarrow \sum_{k=n+1}^m \|f_k\|_{\infty} < \epsilon$. Then for $m \geq n \geq N$, we have

$$\begin{aligned} d_{\infty} \left(\sum_{k=1}^m f_k, \sum_{k=1}^n f_k \right) &= d_{\infty} \left(\sum_{k=n+1}^m f_k, 0 \right) = \left\| \sum_{k=n+1}^m f_k \right\|_{\infty} \\ &\leq \sum_{k=n+1}^m \|f_k\|_{\infty} < \epsilon. \end{aligned}$$

It follows that the sequence of partial sums $(\sum_{n=1}^N f_n)$ is Cauchy. Since $\mathbb{R}$ is complete, so is $C(X \rightarrow \mathbb{R})$, so the limit $f = \sum_{n=1}^{\infty} f_n$ is also in $C(X \rightarrow \mathbb{R})$.

By uniqueness of limits, f is also the point-wise limit of this sequence, so $f(x) = \lim_{n \rightarrow \infty} \sum_{k=1}^n f_k(x) = \sum_{k=1}^{\infty} f_k(x)$.

Example. $\lim_{n \rightarrow \infty} x^n = 0$ for $x \in (-1, 1)$, so f_n converges point-wise to 0. Suppose f_n converges uniformly to a function f . Then $f = 0$ since f_n also converges point-wise and by uniqueness of limits. Let $\epsilon = 1/2$. Suppose there exists $N \in N$ such that $n \geq N \rightarrow |x^n| < 1/2$ for all $x \in (-1, 1)$.

Since x^N is continuous, $\lim_{x \rightarrow 1; x \in (-1, 1)} x^N = 1$. Hence, pick $\delta > 0$ such that $x \in V_{\delta}(1) \cap (-1, 1) \rightarrow x^N \in V_{1/2}(1) = (1/2, 3/2)$. Pick any $x_0 \in V_{\delta}(1) \cap (-1, 1)$, so $x_0^N > 1/2$, contradiction.

4 Uniform Convergence and Integration

Let $a < b$ in $\mathbb{R}$. Let $f_n : [a, b] \rightarrow \mathbb{R}$ be a sequence converging uniformly to f . Then f is RI and

$$\lim_{n \rightarrow \infty} \int_{[a, b]} f_n = \int_{[a, b]} f.$$

Since f_n is bdd, so is f . Let $\epsilon > 0$. Pick $N \in \mathbb{N}$ st $n \geq N \rightarrow d_\infty(f, f_n) < \epsilon$. Fix $n \geq N$. Let $\overline{f_n}$ be pc maj f_n . Then $\overline{f} = \overline{f_n} + \epsilon$ is pc maj f . Then

$$\int_I \overline{f} \leq pc \int_I (\overline{f_n} + \epsilon) = pc \int_I \overline{f_n} + pc \int_I \epsilon = pc \int_I \overline{f_n} + (b-a)\epsilon.$$

Since $\overline{f_n}$ is arbitrary, taking inf, we have $\overline{\int_I f} \leq \int_I f_n + (b-a)\epsilon$. Similarly, $\underline{\int_I f} \geq \int_I f_n - (b-a)\epsilon$.

In particular, for all $\epsilon > 0$,

$$\overline{\int_I f} - \underline{\int_I f} \leq 2(b-a)\epsilon,$$

so $\overline{\int_I f} = \underline{\int_I f}$, ie f is RI.

The previous result gives for all $\epsilon > 0$, there exists $N \in \mathbb{N}$ st $n \geq N$ implies $\int_I f_n \in V_\epsilon(\int_I f)$, ie $\lim_{n \rightarrow \infty} \int_I f_n = \int_I f$ (since $b-a$ is constant).

Let $f_n : I \rightarrow \mathbb{R}$ be RI such that $\sum_{n=1}^\infty f_n$ converges uniformly. Then

$$\int_I \sum_{n=1}^\infty f_n = \lim_{n \rightarrow \infty} \int_I \sum_{k=1}^n f_k = \lim_{n \rightarrow \infty} \sum_{k=1}^n \int_I f_k = \sum_{n=1}^\infty \int_I f_n.$$

5 Uniform Convergence and Derivatives

Consider $f_n : [0, 2\pi] \rightarrow \mathbb{R}$ given by $f_n(x) = \frac{1}{\sqrt{n}} \sin nx$. Let $f : [0, 2\pi] \rightarrow \mathbb{R}$ be the zero function. $d_\infty(f_n, f) = 1/\sqrt{n}$. Since $1/\sqrt{n} \rightarrow 0$ as $n \rightarrow \infty$, the f_n converge uniformly to f . The f_n are differentiable and $f'_n = n^{1/2} \cos nx$, which does not converge to $f' = 0$.

Now consider $f_n : [-1, 1] \rightarrow \mathbb{R}$ given by $f_n(x) = \sqrt{1/n^2 + x^2}$. f_n is differentiable by the composition $x \mapsto 1/n^2 + x^2 \mapsto \sqrt{1/n^2 + x^2}$ (since $1/n^2 + x^2 > 0$ avoids a cusp). $|x| \leq f_n(x) \leq |x| + 1/n$.

Suppose $f_n : [a, b] \rightarrow \mathbb{R}$ is a sequence of continuously differentiable functions such that f'_n converges uniformly to $g : [a, b] \rightarrow \mathbb{R}$. Suppose further there exists $x_0 \in [a, b]$ such that $\lim_{n \rightarrow \infty} f_n(x_0) = L$ exists. Then there exists a differentiable function f such that f_n converges uniformly to f and $f' = g$.

$f'_n : [a, b] \rightarrow \mathbb{R}$ is continuous on a compact set and therefore bdd. It follows $g \in C([a, b] \rightarrow \mathbb{R})$.

Let $f : [a, b] \rightarrow \mathbb{R}$ be given by $f(x) = L - \int_{[a, x_0]} g + \int_{[a, x]} g$, which is well-defined since $g \in C([a, b] \rightarrow \mathbb{R})$ and therefore RI. Since g is continuous, $f' = g$.

I claim that the f_n converge uniformly to f .

Let $\epsilon > 0$. Pick $N \in \mathbb{N}$ st $n \geq N \rightarrow d_\infty(f_n, g) \leq \epsilon$ and $|f_n(x_0) - L| \leq \epsilon$. Then for all $x \in [a, b]$, $f_n(x) - \epsilon \leq g(x) \leq f_n(x) + \epsilon$, so for $x \leq x_0$,

$$\begin{aligned} (x_0 - a)\epsilon &\geq (x_0 - x)\epsilon \geq \left| \int_{[x, x_0]} g - f'_n \right| = |f(x_0) - f(x) - f_n(x_0) + f_n(x)| \\ &= |-f(x) + f_n(x) + L - f_n(x_0)| \geq |-f(x) + f_n(x)| - |L - f_n(x_0)|, \end{aligned}$$

so $|f(x) - f_n(x)| \leq (x_0 - a + 1)\epsilon$.

For $x \geq x_0$,

$$\begin{aligned} (b - x_0)\epsilon &\geq (x - x_0)\epsilon \geq \left| \int_{[x_0, x]} g - f'_n \right| = |f(x) - f(x_0) - f_n(x) + f_n(x_0)| \\ &= |f(x) - f_n(x) + (f_n(x_0) - L)| \geq |f(x) - f_n(x)| - |f_n(x_0) - L|, \end{aligned}$$

so $|f(x) - f_n(x)| \leq (b - x_0 + 1)\epsilon$. Since $x_0 - a + 1, b - x_0 + 1$ are constant, the f_n converge uniformly to f .

Suppose f_n is a sequence of continuously differentiable functions such that $\sum_{n=1}^\infty \|f'_n\|_\infty$ converges. Then $(\sum_{n=1}^N f'_n)$ is a sequence of continuous functions converging uniformly to a function g . Suppose there exists $x_0 \in [a, b]$ such that $\sum_{n=1}^\infty f_n(x_0) = \lim_{N \rightarrow \infty} \sum_{n=1}^N f_n(x_0)$ exists. Then $\sum_{n=1}^\infty f_n$ is differentiable with derivative $\sum_{n=1}^\infty f'_n$.

Suppose $f_n : [a, b] \rightarrow \mathbb{R}$ are differentiable and f'_n converge uniformly to g . Suppose there exists $x_0 \in [a, b]$ such that $\lim_{n \rightarrow \infty} f_n(x_0)$ exists. Then f_n converge uniformly to a differentiable function f such that $f' = g$.

Since (f'_n) is convergent, (f'_n) is Cauchy. Since $(f_n(x_0))$ is convergent, $(f_n(x_0))$ is Cauchy. Let $\epsilon > 0$ and pick $N \in \mathbb{N}$ st $n, m \geq N \rightarrow d_\infty(f'_n, f'_m) \leq \epsilon$ and $|f_n(x_0) - f_m(x_0)| \leq \epsilon$.

I claim that for all $x \in [a, b]$, we have $|(f_n(x) - f_m(x)) - (f_n(x_0) - f_m(x_0))| \leq \epsilon|x - x_0|$. If $x = x_0$, the result holds. Else, suppose $x < x_0$. Define $g : [x, x_0] \rightarrow \mathbb{R}$ by $g(t) = f_n(t) - f_m(t)$. Since f_n, f_m are diff on $[x, x_0]$, by MVT there exists $c \in (x, x_0)$ such that

$$\frac{(f_n(x) - f_m(x)) - (f_n(x_0) - f_m(x_0))}{x - x_0} = f'_n(c) - f'_m(c).$$

The result holds similarly when $x > x_0$. Then by Triangle inequality, $|f_n(x) - f_m(x)| \leq \epsilon|x - x_0| + |f_n(x_0) - f_m(x_0)| \leq \epsilon(b - a + 1)$, so (f_n) is Cauchy. Since $\mathbb{R}$ is complete, $C(X \rightarrow \mathbb{R})$ is complete, so (f_n) converges uniformly to a function f .

Let $\epsilon > 0$. Since the f_n converge uniformly to f , let $n \in \mathbb{N}$ such that $d_\infty(f_n, f) \leq \epsilon$ and $d_\infty(f'_n, g) \leq \epsilon$. Then $|f(x) - f(x_1) - (f_n(x) - f_n(x_1))| \leq |f(x) - f_n(x)| + |f(x_1) - f_n(x_1)| \leq 2\epsilon$. $|f_n(x) - f_n(x_1) - f'_n(x_1)(x - x_1)| \leq \epsilon|x - x_1|$. $|g(x_1) - f'_n(x_1)| \leq \epsilon$, so $|f_n(x) - f_n(x_1) - g(x_1)(x - x_1)| \leq 2\epsilon|x - x_1| + \epsilon|x - x_1| = 3\epsilon|x - x_1|$.

6 Uniform Approximation by Polynomials

Let $f : \mathbb{R} \rightarrow \mathbb{R}$. Let $\text{support } f = \overline{f^{-1}(\mathbb{R} - 0)}$, so $\text{support } f$ is closed by definition. Suppose $\text{support } f$ is compact. Then $f^{-1}(\mathbb{R} - 0)$ is bounded, say by $[a, b]$. For $x \notin [a, b]$, $f(x) = 0$. Conversely, suppose $f^{-1}(\mathbb{R} - 0) \subset [a, b]$ for some $a < b$ in $\mathbb{R}$. Then $\text{support } f \subset [a, b]$ is compact.

Suppose f is compactly supported. Define $\int_{-\infty}^{\infty} f = \int_{[a, b]} f$, where $\text{support } f \subset [a, b]$.

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and supported on $[a, b]$ and $[c, d]$. Then $\int_{[a, b]} f = \int_{[c, d]} f$.

Proof: $f^{-1}(\mathbb{R} - 0) \subset [a, b] \cap [c, d]$. $[a, b] \cap [c, d]$ is a closed interval, say $[u, v]$.

Observe that $f^{-1}(\mathbb{R} - 0) \subset [u, v] \subset [a, b]$. By Theorem 11.4.1(g), we have $\int_{[u, v]} f = \int_{[a, b]} f$. Similarly, $\int_{[u, v]} f = \int_{[c, d]} f$, so $\int_{[a, b]} f = \int_{[c, d]} f$.

Polynomials can approximate the identity. For all $\epsilon > 0$ and $\delta \in (0, 1]$, there exists a polynomial (ϵ, δ) approximation to the identity.

Proof: Let $n \in \mathbb{Z}^+$. Let $A = \int_{-1}^1 (1 - x^2)^n dx$ and $p : [-1, 1] \rightarrow \mathbb{R}$ be given by $p(x) = \frac{1}{A}(1 - x^2)^n$. Observe that

$$A = \int_{-1}^1 (1 - x^2)^n dx \geq 2 \int_0^{1/\sqrt{n}} (1 - x^2)^n dx \geq 2 \int_0^{1/\sqrt{n}} (1 - nx^2) dx = \frac{2}{3\sqrt{n}}.$$

Observe that $1 - \delta^4 \in [0, 1]$. Pick $n \in \mathbb{Z}^+$ such that $(1 - \delta^4)^n < \frac{2\epsilon\delta^2}{3}$. Then

$$(1 - \delta^4)^n = (1 - \delta^2)^n(1 + \delta^2)^n \geq (1 - \delta^2)^n n \delta^2 \geq \sqrt{n}(1 - \delta^2)^n \delta^2,$$

so $\frac{3\sqrt{n}}{2}(1 - \delta^2)^n < \epsilon$. That is, $p(\delta) < \epsilon$. Since p is increasing on $[-1, -\delta]$ and decreasing on $[\delta, 1]$, p is an (ϵ, δ) approximation to the identity.

Suppose $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are continuous compactly supported functions. Let $f * g : \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$(f * g)(x) = \int_{-\infty}^{\infty} f(y)g(x - y)dy.$$

$f * g$ is well-defined. Fix $x \in \mathbb{R}$. Then $f(y)g(x-y)$ viewed as a function of y is continuous and compactly supported.

$$\begin{aligned}(f * g)(x) &= \int_{-\infty}^{\infty} f(y)g(x-y)dy = \int_a^b f(y)g(x-y)dy = \int_{x-b}^{x-a} f(x-y)g(y)dy \\ &= \int_{-\infty}^{\infty} f(x-y)g(y)dy = (g * f)(x).\end{aligned}$$

$$\begin{aligned}((f+g) * h)(x) &= \int_{-\infty}^{\infty} (f+g)(y)h(x-y)dy = \int_{-\infty}^{\infty} (f(y)+g(y))h(x-y)dy \\ &= \int_{-\infty}^{\infty} (f(y)h(x-y) + g(y)h(x-y))dy = \int_{-\infty}^{\infty} f(y)h(x-y)dy + \int_{-\infty}^{\infty} g(y)h(x-y)dy \\ &= (f * h)(x) + (g * h)(x).\end{aligned}$$

Let g be an (ϵ, δ) approximation to the identity. Then $1 - 2\epsilon \leq \int_{-\delta}^{\delta} g(x)dx \leq 1$.

Proof: Since g is nonnegative, $\int_{-\delta}^{\delta} g(x)dx \leq \int_{-1}^1 g(x)dx = 1$. Since g is bounded above by ϵ on $[-1, -\delta], [\delta, 1]$, we have

$$\int_{-\delta}^{\delta} g(x)dx = 1 - \int_{-1}^{-\delta} g(x)dx - \int_{\delta}^1 g(x)dx \geq 1 - 2(1-\delta)\epsilon \geq 1 - 2\epsilon.$$

Let f be supported on $[0, 1]$ and be bounded by M . Let g be an (ϵ, δ) approximation to the identity. Then $|f * g(x) - f(x)| \leq (4M + 1)\epsilon$.

Proof: Observe

$$f * g(x) = \int_{-1}^1 f(x-y)g(y)dy = \int_{-\delta}^{\delta} f(x-y)g(y)dy + \int_{-1}^{-\delta} f(x-y)g(y)dy + \int_{\delta}^1 f(x-y)g(y)dy.$$

$$\int_{-\delta}^{\delta} f(x-y)g(y)dy \leq \int_{-\delta}^{\delta} (f(x) + \epsilon)g(y)dy = (f(x) + \epsilon) \int_{-\delta}^{\delta} g(y)dy \leq f(x) + \epsilon.$$

$$\int_{-\delta}^{\delta} f(x-y)g(y)dy \geq \int_{-\delta}^{\delta} (f(x) - \epsilon)g(y)dy \geq (f(x) - \epsilon)(1 - 2\epsilon),$$

as long as $\epsilon < 1/2$.

7 Real Analytic Functions

Let X be a countably infinite set and $f : X \rightarrow \mathbb{R}$ a function such that $\sum_{x \in X} f(x)$ is absolutely convergent. Suppose $X_0 \subset X_1 \subset X_2 \subset \dots$ is an ascending chain

of finite sets such that $\bigcup_{n=0}^{\infty} X_n = X$. Then $\lim_{n \rightarrow \infty} \sum_{x \in X_n} f(x) = \sum_{x \in X} f(x)$.

Proof: Let $a_n = |X_n|$ for $n \in N$. Let $h_0 : [a_0 - 1] \rightarrow X_0$ be a bijection. Suppose a_k is defined. Define the bijection $h_{k+1} : [a_{k+1} - 1] \rightarrow X_{k+1}$ by extending h_k . Extend $h_0, h_1, h_2, \dots$ to $h : N \rightarrow X$. I claim h is bijective.

Suppose $h(n) = h(n')$ for some $n, n' \in N$. The sequence (a_k) is unbounded since $\bigcup_{n=0}^{\infty} X_n = X$ is infinite. Pick k such that $n, n' \in [a_k - 1]$. Then $h_k(n) = h_k(n')$. Since h_k is injective, $n = n'$, so h is injective.

Suppose $x \in X$. Then $x \in X_k$ for some $k \in N$. Since h_k is surjective, pick $n \in N$ such that $h(n) = h_k(n) = x$, so h is surjective.

Observe that $\sum_{x \in X} f(x) = \sum_{n=0}^{\infty} f(h(n))$. Since the sequence of partial sums $(\sum_{m=0}^n f(h(m)))_{n \in N}$ converges, the subsequence of partial sums $(\sum_{m=0}^{a_n-1} f(h(m)))_{n \in N}$ converges to the same value. In particular, since $h([a_n - 1]) = X_n$,

$$\sum_{m=0}^{a_n-1} f(h(m)) = \sum_{x \in X_n} f(x),$$

so $\lim_{n \rightarrow \infty} \sum_{x \in X_n} f(x) = \sum_{x \in X} f(x)$, as desired.

Let $(a_n), (b_m)$ be nonnegative sequences such that the series $\sum_{n=0}^{\infty} a_n = L$ and $\sum_{m=0}^{\infty} b_m = M$ converge. Then $\sum_{(n,m) \in N^2} a_n b_m = LM$ converges as well.

Proof: Suppose $A \subset N^2$ is finite. Pick $u, v \in N$ such that $A \subset [u] \times [v]$. Then

$$\sum_{(n,m) \in A} a_n b_m \leq \sum_{(n,m) \in [u] \times [v]} a_n b_m = \sum_{n=0}^u a_n \cdot \sum_{m=0}^v b_m \leq LM,$$

so $\sum_{(n,m) \in N^2} a_n b_m$ converges absolutely. Let $X = N^2$ and $X_n = [n] \times [n]$. Applying the previous lemma,

$$\begin{aligned} \sum_{(n,m) \in N^2} a_n b_m &= \lim_{N \rightarrow \infty} \sum_{(n,m) \in [N] \times [N]} a_n b_m = \lim_{N \rightarrow \infty} \sum_{n=0}^N a_n \cdot \sum_{m=0}^N b_m \\ &= \lim_{N \rightarrow \infty} \sum_{n=0}^N a_n \cdot \lim_{N \rightarrow \infty} \sum_{m=0}^N b_m = LM, \end{aligned}$$

as desired.

Corr: Suppose X, Y are sets, $f : X \rightarrow \mathbb{R}, g : Y \rightarrow \mathbb{R}$ are functions such that $\sum_{x \in X} f(x), \sum_{y \in Y} g(y)$ are absolutely convergent. Then $\sum_{(x,y) \in X \times Y} f(x)g(y) = \sum_{x \in X} f(x) \cdot \sum_{y \in Y} g(y)$ is absolutely convergent.

Suppose $f : X \rightarrow \mathbb{R}, g : Y \rightarrow \mathbb{R}$ are such that $X, Y \subset \mathbb{R}$ and $x_0 \in (X \cap Y)^\circ$. Suppose further f, g are analytic at x_0 . Then $fg : X \cap Y \rightarrow \mathbb{R}$ is analytic at x_0 .

Proof: Pick $r > 0$ such that $V_r(x_0) \subset X \cap Y$ and that r is at most the radius of convergence wrt f, g at x_0 . Then $f(x) = \sum_{n=0}^{\infty} c_n(x - x_0)^n$ and $g(x) = \sum_{m=0}^{\infty} d_m(x - x_0)^m$ converge absolutely for $x \in V_r(x_0)$. In particular,

$$f(x)g(x) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n c_k d_{n-k} \right) (x - x_0)^n$$

converges absolutely for $x \in V_r(x_0)$.

Let $f \in C^\infty(\mathbb{R}, \mathbb{R})$. Suppose there exists $x_0 \in \mathbb{R}$ and a sequence (x_n) such that $x_n \neq x_0$, $\lim_{n \rightarrow \infty} x_n = x_0$, and $f(x_n) = 0$. Then $f^{(k)}(x_0) = 0$ for all $k \in \mathbb{N}$.

Proof: I proceed by induction. Suppose $f^{(k)}(x_n) = 0$ for some sequence (x_n) such that $x_n \neq x_0$ and $x_n \rightarrow x_0$. I claim that $f^{(k)}(x_0) = 0$ and there exists a sequence (y_n) such that $y_n \neq x_0$, $y_n \rightarrow x_0$, and $f^{(k+1)}(y_n) = 0$.

Since $f \in C^\infty(\mathbb{R}, \mathbb{R})$, $f \in C^{k+1}(\mathbb{R}, \mathbb{R})$, so $f^{(k)}$ is differentiable and therefore continuous. It follows $f^{(k)}(x_0) = \lim_{n \rightarrow \infty} f^{(k)}(x_n) = \lim_{n \rightarrow \infty} 0 = 0$.

Let $n \in \mathbb{N}$. Suppose $x_n < x_0$. Since $f^{(k)}$ is differentiable on $[x_n, x_0]$ and $f^{(k)}(x_n) = f^{(k)}(x_0)$, by Rolle's theorem there exists $y_n \in (x_n, x_0)$ such that $f^{(k+1)}(y_n) = 0$. The case is analogous when $x_n > x_0$. By Squeeze lemma, the sequence (y_n) converges to x_0 . By assumption, $y_n \neq x_0$.

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is analytic and there exists $x_0 \in \mathbb{R}$ and a sequence (x_n) such that $x_n \neq x_0$, $x_n \rightarrow x_0$, and $f(x_n) = 0$. Then $f \equiv 0$.

First a lemma.

Lemma: Let $f : E \rightarrow \mathbb{R}$ be analytic at $x_0 \in E^\circ$. Let $r > 0$ be such that $V_r(x_0) \subset E$ and $2r$ is at most the radius of convergence at x_0 . Suppose there exists $x_1 \in V_r(x_0)$ such that $f^{(k)}(x_1) = 0$ for all $k \in \mathbb{N}$. Then f is identically zero on $V_r(x_0)$.

Let $x \in V_r(x_0)$ and $\Delta x = x - x_0$. Observe that

$$\sum_{n=0}^{\infty} \frac{|\Delta x|^n}{n!} \sum_{k=n}^{\infty} \left(c_k \prod_{j=1}^n (k - j + 1) \right) |\Delta x|^{k-n} = \sum_{n=0}^{\infty} \sum_{k=n}^{\infty} c_k \binom{k}{n} |\Delta x|^k \quad (1).$$

Observe that

$$\sum_{n=0}^{\infty} c_n \left(\sum_{k=0}^n \binom{n}{k} \right) |\Delta x|^n = \sum_{n=0}^{\infty} c_n |2\Delta x|^n,$$

which converges absolutely since r is at most $1/2$ of the radius of convergence at x_0 . It follows that (1) converges. Consider the sum

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{(-1)^n (\Delta x)^n}{n!} \sum_{k=n}^{\infty} \left(c_k \prod_{j=1}^n (k-j+1) \right) (\Delta x)^{k-n} &= \sum_{n=0}^{\infty} \sum_{k=n}^{\infty} c_k (-1)^n \binom{k}{n} (\Delta x)^k \\ &= \sum_{n=0}^{\infty} c_n \left(\sum_{k=0}^n (-1)^k \binom{n}{k} \right) (\Delta x)^n = c_0. \end{aligned}$$

Take $x = x_1$. Since $f^{(n)}(x_1) = \sum_{k=n}^{\infty} \prod_{j=1}^n (k-j+1) (\Delta x_1)^{k-n} = 0$, we have $c_0 = 0$. In general, differentiating k times (since f is infinitely differentiable at x_0), we have $k!c_k = 0$, so $c_k = 0$ for all $k \in N$. It follows f is identically zero on $V_r(x_0)$, proving the lemma.

Since $f : \mathbb{R} \rightarrow \mathbb{R}$ is analytic, $f \in C^\infty(\mathbb{R}, \mathbb{R})$. By two lemmas before, $f^{(k)}(x_0) = 0$ for all $k \in N$. Let $a = x_0$ and fix $b > 0$. Cover the interval $[a, b]$ with open intervals $V_r(x_1)$ and denote this set by $\mathcal{A}$, where $x_1 \in [a, b]$ and r is defined as before. $\mathcal{A}$ is indeed a cover since f is analytic.

By Lebesgue number lemma, partition the interval into $0 = x_0 < x_1 < x_2 < \dots < x_n = b$ such that $[x_i, x_{i+1}] \subset A$ for some $A \in \mathcal{A}$ for all $i \in [[n-1]]$. I claim that f is identically zero on $[a, b]$.

I proceed by induction. Assume $f^{(k)}(x_i) = 0$ for all $k \in N$ for some $i \in [[n-1]]$. There exists $c \in \mathbb{R}$ and $r > 0$ such that $[x_i, x_{i+1}] \subset V_r(c)$. By the previous lemma, f is identically zero on $V_r(c)$ and therefore $[x_i, x_{i+1}]$. In particular, $f^{(k)}(x_{i+1}) = 0$ for all $k \in N$. It follows f is identically zero on $[a, b]$. Since this holds for any $b > a$, f is identically zero on $[a, \infty)$. A similar argument shows that f is identically zero on $(-\infty, a]$, so $f \equiv 0$ on $\mathbb{R}$.

Sps $f : V_r(a) \rightarrow \mathbb{R}$ is real analytic at a with radius r and $b \in V_r(a)$, $s > 0$ such that $V_s(b) \subset V_r(a)$. Show that f is real analytic at b with radius s .

Write for $x \in V_s(b)$,

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} c_n (x-a)^n = \sum_{n=0}^{\infty} c_n (x-a+b-b)^n \\ &= \sum_{n=0}^{\infty} c_n \sum_{k=0}^n \binom{n}{k} (x-b)^k (b-a)^{n-k}. \end{aligned}$$

Consider

$$\sum_{n=0}^{\infty} |c_n| \sum_{k=0}^{\infty} \binom{n}{k} |x-b|^k |b-a|^{n-k} = \sum_{n=0}^{\infty} |c_n| (|x-b| + |b-a|)^n \quad (1)$$

$V_s(b) \subset V_r(a) \rightarrow b+s \leq a+r, b-s \geq a-r$, so $a-b \geq -(r-s)$ and $a-b \leq r-s$, so $|a-b| \leq r-s$. Then $|x-b| + |b-a| < s + |b-a| \leq r$. Since $\sum_{n=0}^{\infty} c_n (x-a)^n$ is absolutely convergent for $x \in V_r(a)$, we have $\sum_{n=0}^{\infty} |c_n| \cdot |\Delta x|^n$ is convergent for $|\Delta x| < r$, so (1) is convergent.

Now we freely manipulate:

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} c_n \sum_{k=0}^n \binom{n}{k} (x-b)^k (b-a)^{n-k} = \sum_{n=0}^{\infty} \sum_{k=0}^n c_n \binom{n}{k} (x-b)^k (b-a)^{n-k} \\ &= \sum_{k=0}^{\infty} \sum_{n=k}^{\infty} c_n \binom{n}{k} (x-b)^k (b-a)^{n-k} = \sum_{k=0}^{\infty} \left(\sum_{n=k}^{\infty} c_n \binom{n}{k} (b-a)^{n-k} \right) (x-b)^k, \end{aligned}$$

so f is real analytic at b with radius s .

Suppose $a_n \rightarrow A$ and $b_n \rightarrow B$ are convergent and $\sum_{n=0}^{\infty} (a_{n+1} - a_n) b_n$ is convergent. Then $\sum_{n=0}^{\infty} a_{n+1} (b_{n+1} - b_n)$ is also convergent and $\sum_{n=0}^{\infty} (a_{n+1} - a_n) b_n + \sum_{n=0}^{\infty} a_{n+1} (b_{n+1} - b_n) = AB - a_0 b_0$.

Proof: Observe that

$$\begin{aligned} \sum_{n=0}^{\infty} a_{n+1} b_{n+1} - a_n b_n &= \lim_{N \rightarrow \infty} \sum_{n=0}^N a_{n+1} b_{n+1} - a_n b_n = \lim_{N \rightarrow \infty} a_{N+1} b_{N+1} - a_0 b_0 \\ &= AB - a_0 b_0. \end{aligned}$$

Then

$$\begin{aligned} \sum_{n=0}^{\infty} a_{n+1} (b_{n+1} - b_n) &= \sum_{n=0}^{\infty} a_{n+1} b_{n+1} - a_n b_n - (a_{n+1} - a_n) b_n \\ &= \sum_{n=0}^{\infty} a_{n+1} b_{n+1} - a_n b_n - \sum_{n=0}^{\infty} (a_{n+1} - a_n) b_n = AB - a_0 b_0 - \sum_{n=0}^{\infty} (a_{n+1} - a_n) b_n, \end{aligned}$$

as desired.

Let f be real analytic at a with radius $0 < R < \infty$. Suppose $\sum_{n=0}^{\infty} c_n R^n$ converges. Then $f : (a-R, a+R] \rightarrow \mathbb{R}$ is continuous at $a+R$, ie $\lim_{x \rightarrow a+R; x \in V_R(a)} \sum_{n=0}^{\infty} c_n (x-a)^n = \sum_{n=0}^{\infty} c_n R^n$.

Proof: Let $y = (x-a)/R$ and $d_n = c_n R^n$, so assume f is real analytic at 0 with radius 1 and $\sum_{n=0}^{\infty} d_n$ converges. Then $f : (-1, 1] \rightarrow \mathbb{R}$ is continuous at

1, ie $\lim_{y \rightarrow 1; y \in (-1, 1)} \sum_{n=0}^{\infty} d_n y^n = \sum_{n=0}^{\infty} d_n$.

Let $D = \sum_{n=0}^{\infty} d_n$ and $S_N = \sum_{n=0}^{N-1} d_n - D$, so $S_0 = -D$ and $\lim_{N \rightarrow \infty} S_N = 0$, $S_{N+1} - S_N = d_N$.

Write

$$\sum_{n=0}^{\infty} d_n y^n = \sum_{n=0}^{\infty} (S_{n+1} - S_n) y^n.$$

$\lim_{n \rightarrow \infty} S_n = 0$ and $\lim_{n \rightarrow \infty} y^n = 0$ (since $y \in (-1, 1)$). LHS is assumed to exist, so

$$\begin{aligned} \sum_{n=0}^{\infty} S_{n+1} (y^{n+1} - y^n) &= 0 \cdot 0 - S_0 \cdot 1 - \sum_{n=0}^{\infty} (S_{n+1} - S_n) y^n \\ &= D - \sum_{n=0}^{\infty} (S_{n+1} - S_n) y^n, \end{aligned}$$

so it suffices to show $\lim_{y \rightarrow 1; y \in (-1, 1)} \sum_{n=0}^{\infty} S_{n+1} y^n (1 - y) = 0$.

Let $\epsilon > 0$. Pick $N \in \mathbb{Z}^+$ st $|S_n| < \epsilon$ for $n \geq N$. Then

$$\sum_{n=0}^{\infty} S_{n+1} y^n (1 - y) = \sum_{n=0}^{N-1} S_{n+1} y^n (1 - y) + \sum_{n=N}^{\infty} S_{n+1} y^n (1 - y).$$

Note the right hand term $\sum_{n=N}^{\infty} S_{n+1} y^n (1 - y)$ converges absolutely since

$$\begin{aligned} \sum_{n=N}^{\infty} |S_{n+1}| y^n (1 - y) &\leq \sum_{n=N}^{\infty} \epsilon y^n (1 - y) = \epsilon \sum_{n=N}^{\infty} y^n (1 - y) \\ &\leq \epsilon \sum_{n=0}^{\infty} y^n (1 - y) = \epsilon, \end{aligned}$$

for $y \in (0, 1)$ (restrict to $y > 0$), and in fact

$$\left| \sum_{n=N}^{\infty} S_{n+1} y^n (1 - y) \right| \leq \epsilon.$$

Then

$$\begin{aligned} \left| \sum_{n=0}^{\infty} S_{n+1} y^n (1 - y) \right| &\leq \left| \sum_{n=0}^{N-1} S_{n+1} y^n (1 - y) \right| + \epsilon \\ &\leq \left(\sum_{n=0}^{N-1} |S_{n+1}| \right) \cdot (1 - y) + \epsilon \end{aligned}$$

for all $y \in (0, 1)$, so pick $y \in (1 - \delta, 1)$ such that $\left(\sum_{n=0}^{N-1} |S_{n+1}|\right) \cdot \delta < \epsilon$, so $\lim_{y \rightarrow 1; y \in (-1, 1)} \sum_{n=0}^{\infty} S_{n+1} y^n (1 - y) = 0$, as desired.